PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

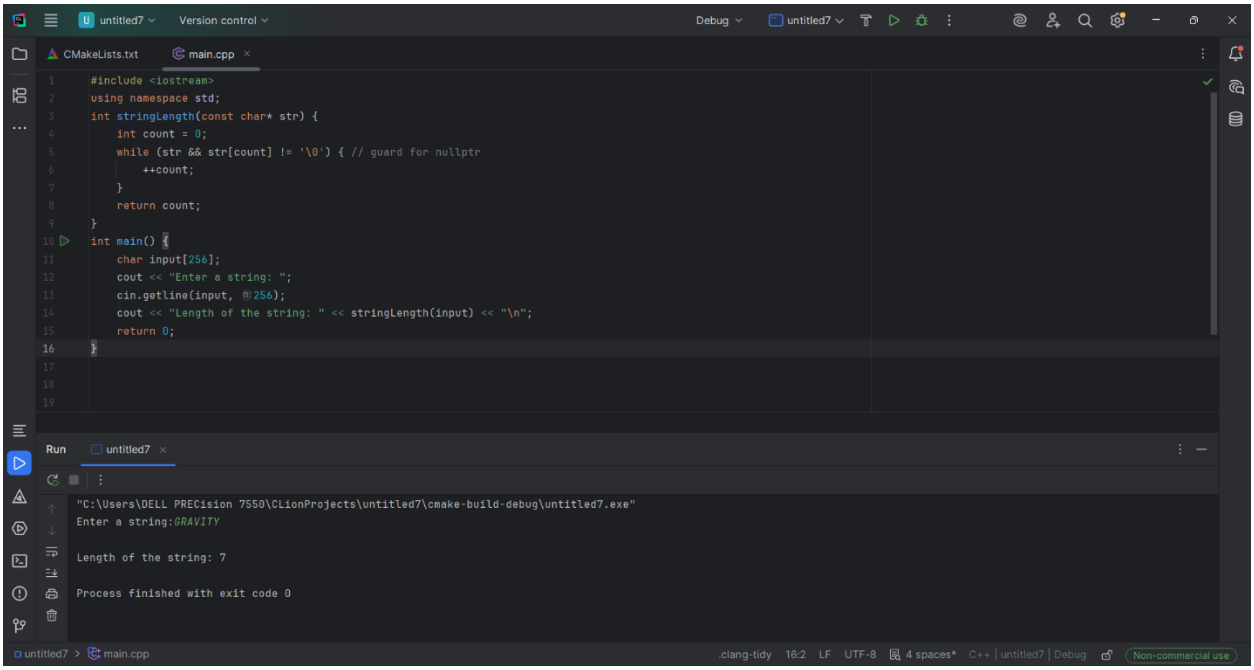
Programming Fundamentals Lab 12	**Instructor: Mr. M Hamza**
Program: AI Fall 2025	**Date: 15 JANUARY, 2025**
Course Code: COMP-111L	

Lab Task

Tasks 1.

Write a C++ program to write the following contents in the file and then display the file. The delimiter is $.

Contents of names2.txt: Jayne Murphy$47 Jones Circle$Almond, NC 28702\n$Bobbie Smith$ 217 Halifax Drive$Canton, NC 28716\n$Bill Hammet$PO Box 121$ Springfield, NC 28357\n$



Task 2.

Iterative Factorial Write an iterative version (using a loop instead of recursion) of the factorial function. Test it with a driver program.

```
#include <iostream>
#include <cstring> // for strlen
using namespace std;
void displayBackward(const char* str) {
    if (!str) return;
    int len = static_cast<int>(strlen(str));
    for (int i = len - 1; i >= 0; --i) {
        cout << str[i];
    }
    cout << "\n";
}
int main() {
    char input[256];
    cout << "Enter a string: ";
    cin.getline(input, 256);
    cout << "Backward: ";
    displayBackward(input);
    return 0;
}
```

```
"C:\Users\DELL PRECision 7550\CLionProjects\untitled7\cmake-build-debug\untitled7.exe"
Enter a string:GRAVITY

Backward: YTIVARG

Process finished with exit code 0
```

Task 3.

Recursive Array Sum Write a function that accepts an array of integers and a number indicating the number of elements as arguments. The function should recursively calculate the sum of all the numbers in the array. Demonstrate the function in a driver program.

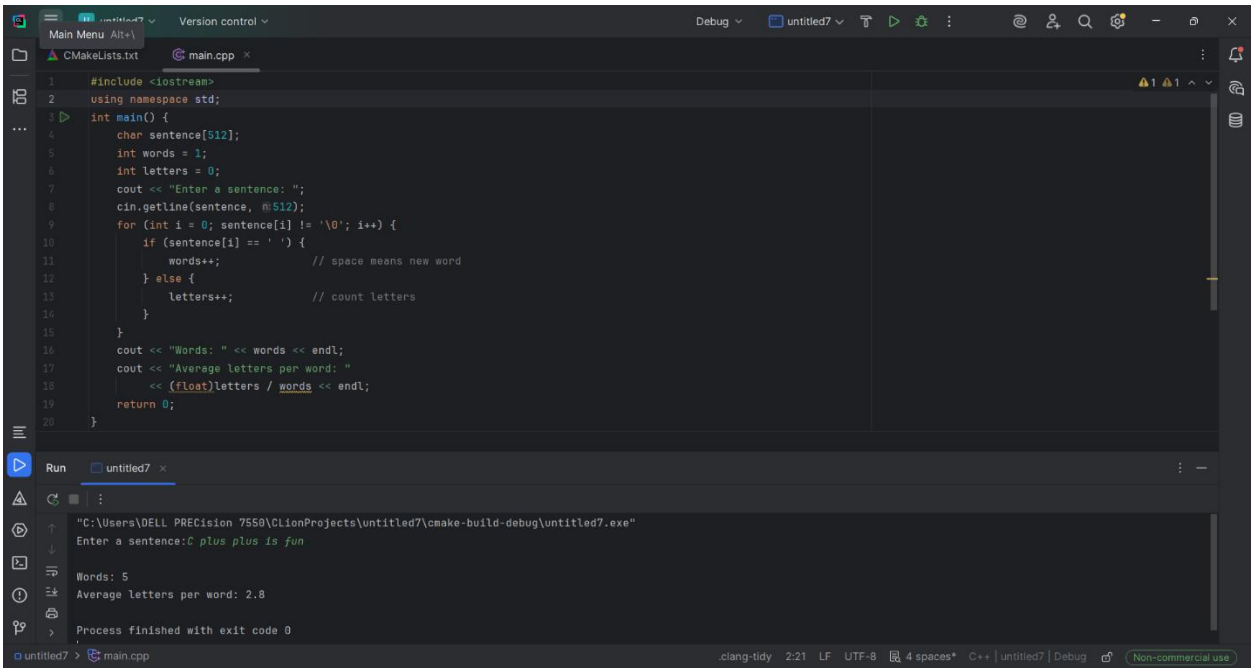
```
#include <iostream>
using namespace std;
int main() {
    char sentence[512];
    int words = 1;
    cout << "Enter a sentence: ";
    cin.getline(sentence, 512);
    // Count spaces
    for (int i = 0; sentence[i] != '\0'; i++) {
        if (sentence[i] == ' ') {
            words++;
        }
    }
    cout << "Number of words: " << words << endl;
    return 0;
}
```

```
"C:\Users\DELL PRECision 7550\CLionProjects\untitled7\cmake-build-debug\untitled7.exe"
Enter a sentence: Four score and seven years ago

Number of words: 7

Process finished with exit code 0
```

Task 4.

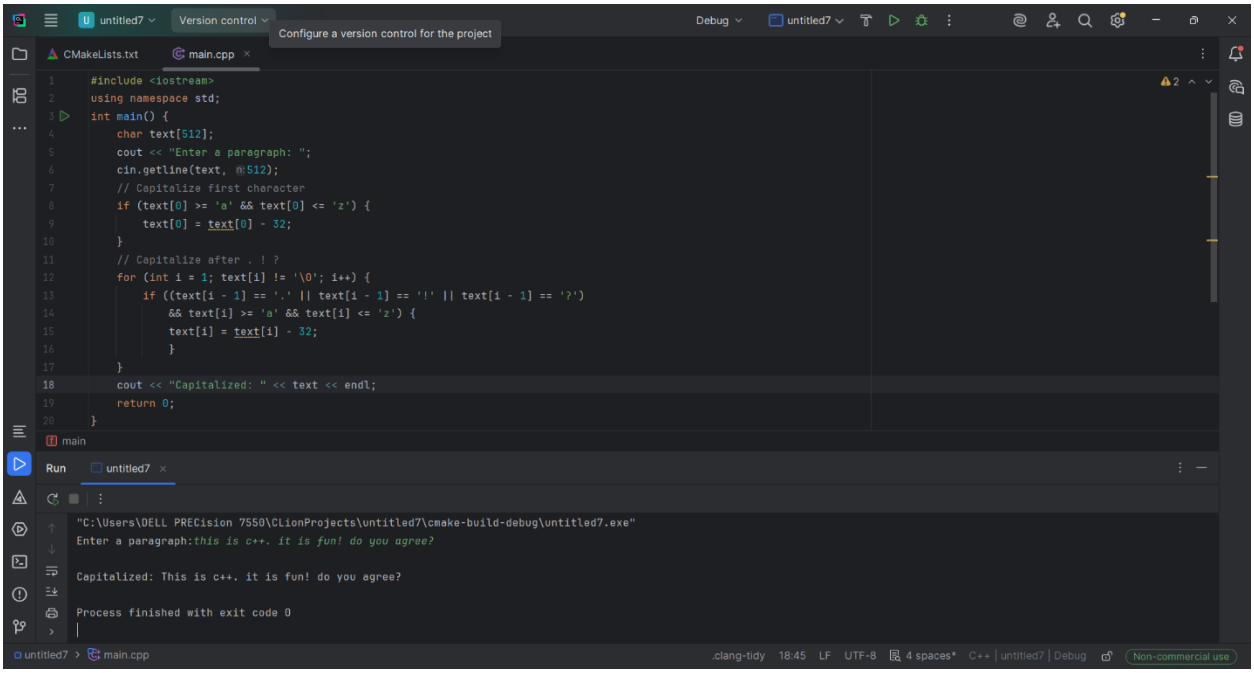
Recursive Multiplication Write a recursive function that accepts two arguments into the parameters x and y. The function should return the value of x times y. Remember, multiplication can be performed as repeated addition:

7*4=4+4+4+ 4+4+4+4 5.



Task 5.

Recursive Power Function Write a function that uses recursion to raise a number to a power. The function should accept two arguments: the number to be raised and the exponent. Assume that the exponent is a nonnegative integer. Demonstrate the function in a program.


untitled7
Version control
Configure a version control for the project
Debug
untitled7
CMakeLists.txt
main.cpp
#include <iostream>
using namespace std;
int main() {
char text[512];
cout << "Enter a paragraph: ";
cin.getline(text, n:512);
// Capitalize first character
if (text[0] >= 'a' && text[0] <= 'z') {
text[0] = text[0] - 32;
}
// Capitalize after . ! ?
for (int i = 1; text[i] != '\0'; i++) {
if ((text[i - 1] == '.' || text[i - 1] == '!' || text[i - 1] == '?')
&& text[i] >= 'a' && text[i] <= 'z') {
text[i] = text[i] - 32;
}
}
cout << "Capitalized: " << text << endl;
return 0;
}
main
Run
untitled7
"C:\Users\DELL PRECision 7550\CLionProjects\untitled7\cmake-build-debug\untitled7.exe"
Enter a paragraph:this is c++. it is fun! do you agree?
Capitalized: This is c++. it is fun! do you agree?
Process finished with exit code 0
untitled7 > main.cpp
.clang-tidy
18:45
LF
UTF-8
4 spaces*
C++ | untitled7 | Debug
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